

AI-Powered Smart Hospital Appointment & Emergency Triage System

Dynamic, risk-aware scheduling that lets a hospital absorb walk-in emergencies without breaking a single booked patient's promise.

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1. Problem Overview

Hospital OPD schedules are built days in advance, yet real hospital floors are interrupted constantly by emergency arrivals. Every time an emergency patient walks in, the doctor treating them goes off-schedule, and every advance-booked patient waiting for that doctor is silently delayed. Reception staff currently juggle this by phone and gut feeling, which causes long waiting-room delays, missed high-risk patients, and no audit trail of who was moved, why, and to whom.

100

Patients in Dataset

88 / 12

Booked vs Emergency

23

Bookings Re-scheduled

0 min

Max Delay After Fix

2. Proposed AI Solution

An automated **Smart Scheduling & Triage Engine** that sits between the hospital's booking system and its doctors' calendars. It continuously scores every patient's vitals, instantly re-routes bookings the moment an emergency lands, and notifies every affected patient in real time — so the front desk never has to make these calls manually.

How it decides:

- **Vitals-based risk score:** heart rate, SpO₂, blood pressure, temperature, respiration rate and consciousness are each checked against danger thresholds; 3+ flags = High risk, 1–2 = Medium, 0 = Low.
- **Senior-first emergency routing:** the most senior free doctor in the matching department is assigned to the emergency, with wait-time scaled to severity (30–60 minutes).
- **Same-time reassignment:** every booked patient caught in that block is moved to another doctor at the *exact same appointment time* — same department first, General Medicine as backup — so nobody is pushed later in the day.
- **Automatic patient messaging:** a confirmation SMS/WhatsApp message is generated the moment a patient is reassigned, naming the new doctor and confirming the unchanged time.

3. Visual Insights from the Pilot Dataset

The two charts below summarise the 100-patient pilot run: how patients arrived, and how well the system absorbed emergencies without creating delays.

Patient Inflow: Advance vs Emergency

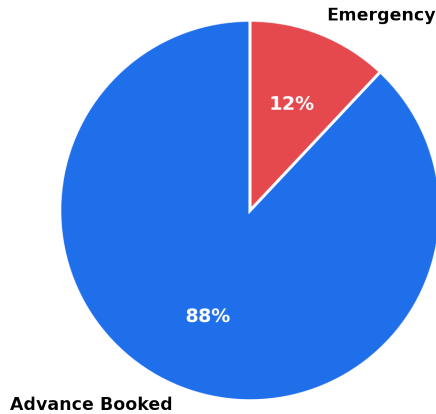


Chart 1 — 88% of patients were pre-booked; 12% arrived as emergencies.

Impact of Emergencies on Booked Schedule

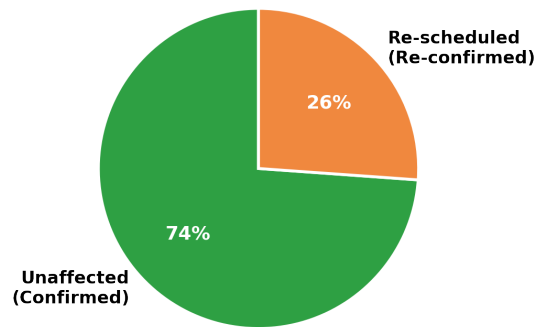


Chart 2 — Even after every emergency, 74% of bookings (65 patients) stayed untouched; the remaining 26% (23 patients) were re-confirmed with a new doctor at the same time.

4. Procedure & Approach

1 Step 1 – Data Intake

Load the daily patient and appointment feed (demographics, vitals, booking type, doctor and time-slot) and run duplicate, missing-value and double-booking checks before anything else runs.

2 Step 2 – Risk Scoring

Score every patient's vitals against clinical thresholds to assign a Low / Medium / High risk tier, so high-risk booked patients can be flagged for a pre-visit nurse call.

3 Step 3 – Schedule Construction

Build the live schedule table from all advance bookings, keeping each patient's original doctor and time as a reference point.

4 Step 4 – Emergency Queue Handling

Process emergency arrivals in order of arrival: assign the most senior free doctor in that department and block their calendar for a severity-scaled duration.

5 Step 5 – Conflict Resolution

For every booked patient caught inside that blocked window, search for a free doctor at the identical time slot — same department first, General Medicine next — and reassign automatically.

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Step 6 – Patient Notification

Generate and log a confirmation message for every reassigned patient, and mark any patient with no free doctor as 'Call Patient' for manual follow-up.

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Step 7 – Validation & Reporting

Re-check the finished schedule for clashes and delays, then generate the pie/bar visual summaries and export the final schedule, messages and emergency log as CSV files for the hospital's records.

5. Expected Outcome

On the 100-patient pilot dataset, the engine handled all 12 emergency arrivals while keeping the day fully on schedule:

Metric	Result
Emergency patients absorbed	12 / 12 (100%)
Booked patients re-routed to a new doctor	23
Booked patients left completely untouched	65
Maximum delay experienced by any patient	0 minutes
Double-bookings or doctor-time clashes after fix	0
Confirmation messages auto-generated	1 per reassigned patient

Broader impact once deployed hospital-wide:

- **Zero silent delays** — patients are moved, never made to wait, because reassignment happens at the same time slot.
- **Safety net for high-risk patients** — automatic flags mean vulnerable patients get a nurse call before they even arrive.
- **Transparent audit trail** — every reassignment, message and emergency event is logged, replacing ad-hoc phone coordination.
- **Lower front-desk load** — staff supervise the system instead of manually renegotiating every doctor's calendar during a crisis.
- **Scalable** — the same rule engine extends cleanly from one OPD floor to an entire multi-department hospital network.

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